A STANDARD COVERING IS REFINED BY A SINGLE FPQC COVERING WITH THE SAME CECH COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the three assertions; the refinement maps in both directions are shown to be inverse, which settles the third.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Standard tau-covering and the big or small site of a scheme. Same chapter, the definition of a standard tau-covering of the Zariski, etale, smooth, syntomic or fppf site of a scheme. Supplies the covering U and the sites in which the statement is made.
- Cech complex of a covering and comparison of Cech complexes under refinement. Same chapter, the Cech complex of a covering and the map induced by a refinement. Supplies the third assertion of the statement and the strategy of comparing the two complexes through refinement maps.

Source reference. Original lemma 020R: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2188–2188. Navigation pointer invoked at proof line 2188; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0219: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2189–2189. Navigation pointer invoked at proof line 2189; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0223: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2190–2190. Navigation pointer invoked at proof line 2190; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0229: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2191–2191. Navigation pointer invoked at proof line 2191; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 021Q: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2192–2192. Navigation pointer invoked at proof line 2192; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original remark 03NY: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 905–905. Navigation pointer invoked at proof line 2197; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0215: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2201–2201. Navigation pointer invoked at proof line 2201; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 021Z: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2202–2202. Navigation pointer invoked at proof line 2202; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0225: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2203–2203. Navigation pointer invoked at proof line 2203; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 021M: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2204–2204. Navigation pointer invoked at proof line 2204; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03F4: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2212–2212. Navigation pointer invoked at proof line 2212; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\tau \in \{fppf, syntomic, smooth, étale, Zariski\}$. Let S be a scheme. Let $\mathcal{F}$ be an abelian sheaf on $(Sch/S)_\tau$, or on S_τ in case $\tau = étale$, and let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be a standard τ-covering of this site. Let $V = \coprod_{i \in I} U_i$. Then*

(1) *V is an affine scheme,*

(2) $\mathcal{V} = \{V \to U\}$ *is an fpqc covering and also a* τ*-covering unless* $\tau = Zariski$,
(3) *the Čech complexes* $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ *and* $\check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{F})$ *agree.*

Proof. The definition of a standard τ-covering is given in Topologies, Definition 020R, 0219, 0223, 0229, and 021Q. By definition each of the schemes U_i is affine and I is a finite set. Hence V is an affine scheme. It is clear that $V \to U$ is flat and surjective, hence $\mathcal{V}$ is an fpqc covering, see Example 03NY. Excepting the Zariski case, the covering $\mathcal{V}$ is also a τ-covering, see Topologies, Definition 0215, 021Z, 0225, and 021M.

Note that $\mathcal{U}$ is a refinement of $\mathcal{V}$ and hence there is a map of Čech complexes $\check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{F}) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$, see Cohomology on Sites, Equation (03F4). Next, we observe that if $T = \coprod_{j \in J} T_j$ is a disjoint union of schemes in the site on which $\mathcal{F}$ is defined then the family of morphisms with fixed target $\{T_j \to T\}_{j \in J}$ is a Zariski covering, and so

$$\mathcal{F}(T) = \mathcal{F}(\coprod\nolimits_{j \in J} T_j) = \prod\nolimits_{j \in J} \mathcal{F}(T_j) \tag{0.1.1}$$

by the sheaf condition of $\mathcal{F}$. This implies the map of Čech complexes above is an isomorphism in each degree because

$$V \times_U \ldots \times_U V = \coprod\nolimits_{i_0, \ldots i_p} U_{i_0} \times_U \ldots \times_U U_{i_p}$$

as schemes. □